

Exercise 2

```
[ec2-user@ip-172-31-20-184 lab8]$ mpicc -o sum_mpi sum_mpi.c
mpirun --oversubscribe -np 4 ./sum_mpi
Rank 3: summed 7500001 to 10000000 => local_sum = 21875001250000
Rank 1: summed 2500001 to 5000000 => local_sum = 9375001250000
Rank 2: summed 5000001 to 7500000 => local_sum = 15625001250000
Rank 0: summed 1 to 2500000 => local_sum = 3125001250000

[Sum] Processes = 4
[Sum] Total sum = 50000005000000
[Sum] Expected = 50000005000000
[Sum] Correct? = YES
[Sum] Time = 0.014428 sec
[ec2-user@ip-172-31-20-184 lab8]$
```

The range 1 to 10,000,000 is divided equally between the processes. Each process works out its own first and last number from its rank ($\text{chunk_size} = N / \text{size}$, the last rank also takes the remainder) and adds them into a long long `local_sum`. No array is needed, so nothing has to be sent to the workers. Every process except root sends its `local_sum` to rank 0 with `MPI_Send`, and root receives them one by one with `MPI_Recv` (`source = r`) and adds them to its own sum.

Total sum = 50000005000000, which matches the expected value $N(N + 1) / 2$ (Correct? = YES). Time = 0.014428 sec with 4 processes.